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Predicting shipment delays with gradient boosting: an explainable approach

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Abstract

We predict shipment delays for 48,210 consignments using XGBoost and interpret the model with SHAP values. The model reaches a 0.81 F1 score on a held-out quarter. We predict shipment delays for 48,210 consignments using XGBoost and interpret the model with SHAP values. The model reaches a 0.81 F1 score on a held-out quarter. We predict shipment delays for 48,210 consignments using XGBoost and interpret the model with SHAP values. The model reaches a 0.81 F1 score on a held-out quarter.

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Introduction

Late deliveries are costly for carriers and customers alike. Prior work relies on linear models. Late deliveries are costly for carriers and customers alike. Prior work relies on linear models. Late deliveries are costly for carriers and customers alike. Prior work relies on linear models. Late deliveries are costly for carriers and customers alike. Prior work relies on linear models. Late deliveries are costly for carriers and customers alike. Prior work relies on linear models.

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Methods

Features include route distance, carrier history and weather. We tune hyperparameters with five-fold cross validation. Features include route

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